

ActInf Livestream #044.1 ~ Therapeutic Alliance as Active Inference: The R

Livestream | Therapeutic Alliance as Active Inference: The Role of Therapeutic Touch.. | 1:47:13

[Music]

hello and welcome everyone

it's may 18 2022 and we are in actin

flab live stream number 44.1 on the

therapeutic alliance as active inference

the role of therapeutic touch and

synchrony

welcome to the active lab we're a

participatory online lab that is

communicating learning and practicing

applied active inference you can find us

at the links on the slide this is a

recorded and an archived live stream so

please provide feedback so we can

improve our work

all backgrounds and perspectives are

welcome and we'll be following video

etiquette for live streams

if you want to learn more about live

streams or any of the other projects

that are happening in the active lab

head over to activeinference.org

okay

today we're in live stream number 44.1

and we are learning and discussing

around this paper therapeutic alliance

as active inference by mcparkin

saratelli fristen and estevus and we're

really honored and appreciative to have

a bunch of authors joining us today and

also ian and ian and i did a zero last

week so for any context

one might want to check out the dot zero

otherwise in the dot one today and in

the dot two next week we're going to

open up and develop threads have the opportunity to address any questions that people are having in the live chat or submitted to us otherwise and just take notes and

have spontaneity and see where this goes because it's a really exciting area of research so we're just going to start with that

um

all say hello and then pass to ian and then we'll hear from the authors with any context so

i'm a researcher in california and

was very interested in this

integration of a lot of perspectives

from like the ecological

and the developmental and kind of

bringing it towards practice which i

hope we can continue the discussion

around and i'll pass it to ian

thank you daniel um so my name's ian i

am based in england i work as a hands-on

manual therapist um have a background in

clinical and surgical sciences but also

do a little bit of research

now around the topic of interception and

this is where

um the work of these authors caught my

attention um the whole series actually

on inactivism and active inference it's

kind of where i've arrived at in the

last year

and i'm interested in

both how we talk about these ideas to

patients um

sort of in service of helping them

um

updates the meaning they sort of give to
their situation they're in
and also from a research point of view
you know
some how how we might be able to drive
forward some new questions in
research around health
thank you over to whoever's next
okay uh so i'm george estevez
i'm based between malta and italy uh i'm
an osteopath a researcher um
so my main interests uh are on
you know the things we're talking about
here today so inactivism active
inference
um
the sort of management of of of chronic
pain and complexity in clinical practice
a bit of background apart from from you
know osteopathy
my phd was exploring the role of
multi-sensory integration in
during palpation uh in osteopathy you
know
several moons ago so that's uh that was
my sort of
early dive into you know this kind of
world
and back in the sort of late 2000s when
people start talking about bayesian
models uh to explain multi-sensory
integration and so on so and that's
where where kind of things start from
hi i'm francesco uh
italian based in italy and thanks daniel
and ian for leading this
uh interesting
uh hopefully interesting live stream um

um
nostropod so handsome treatment and
also a neuroscientist
um so my main interest is in exploring
um effect of manual therapies within
the um
the uh the concept of interception and
the autonomy nervous system effects
that was my uh
basically my my main
streamline in the during the
my
phd
and then with george uh we um we
explored the uh
active inference and activism uh in the
last couple of years
and so uh and we're trying to also apply
in the context of therapeutic alliance
and different in different
um
age groups so starting from
the neonates preterms where actually we
have also
a big
quite a strong background and then into
uh ordered
persistent physical symptoms and also in
the other
age groups so that is the um so i'm just
divided into the clinical part and the
and the research
activities
thank you
sorry
[Music]
hi i'm zoe i'm the primary author on
this paper um again i am a osteopath and

um

this is my first paper that i've
published i originally started doing
a little bit of research on chronic pain
and how it affects autonomic a long
time ago um and then about two years ago
i reconnected with george on his
um active inference and
interceptive ideas
and since then we've kind of
gone from there and um this paper was
published we
i focus as an osthpath i'm mainly touch
based
um and so i wanted to do a paper based
on how touch
affects things and kind of have a little
bit more of the neuroscience behind what
we do
um so that's
and active imprint seemed the perfect
way to try and
justify things that maybe we
as osteopathic manual therapists maybe
feel but haven't been able to put down
um empirically on paper so that's how
the paper kind of started about
awesome
so again anyone in the live chat or
anyone here can just raise their hand or
ask a question but let's just jump in
right there like
that's really interesting about
using active inference for formalizing
some of the intuitions
and experiences that might be
felt during
the clinical setting so

what is the clinical setting
that you are engaged in or focusing on
and then what was that process of
recognizing
some experience and then how does active
inference fit in with formalizing that
[Music]
uh so
actually
you know a little bit of what i
mentioned before the sort of early
journey started with trying to explore
actually what we
what we do as osteopaths you know one of
the things or what we do as manual
therapies one of the things we do is we
first
there's a kind of the big claims about
what what we can do and what we can feel
and almost like having this sort of
magical sort of ability to
to recognize what's wrong with the
patient and that was always based around
this idea that
you know manual therapists or osteopaths
have superior abilities to to feel for
things you know palpation is really the
big thing
and based on on a kind of a paradigm
that's um
you know a little bit of old school
biomedically based biomechanically based
that
you know dysfunction altered mobility
for example equals pain equals problem
and of course
you know as the knowledge in the field
of pain science evolved we we started

realizing that's that's not the case you know because you know what happens with chronic pain is there is no longer a tissue dysfunction in lots of cases but there is pain so clearly it becomes extremely difficult to find things find things that actually are reliable and and that's what the reliability is is an observation and it's been one of the big debates in the manual therapy world which is we cannot reliably find what we claim we can find so poor typically poor interoperator reliability two people cannot agree on things and including also very poor intra operator reliability you know i if i test one thing and i retest that i will not find the same thing and of course you know this is the kind of um you know some some important things to take into account if the patient has a real pathology uh for example uh a real joint restrict restriction of movement for example associated with severe arthritis then you know one should find something you know it's clearly there is is objective but what happens in most of the patients is what we claim to find are things that are very qualitative in nature very subjective so so the initial kind of journey was really to try to understand really what's going on and hence the the idea of exploring things from a kind of multi-sensory perspective uh clearly what we find or what we rely

as in the hands
uh it's not simply the haptic system of
the or you know somatosensory system
is a multi-sensory uh experience you
know we see we observe we touch we
change
we listen
and and that creates straight away
an environment where there's so many uh
variables that becomes extremely hard to
say yeah for sure
this is what i find so you know you you
can have people criticizing for this on
the sake of
the sake of criticizing or saying
everything is a kind of a it's an
illusion
to other people claiming no it's for
sure it's true and i think you know we
need to understand that actually
probably we sit in in between and more
than 10 years ago now i remember when i
kind of started presenting some of the
you know some of the research findings
to the to my profession
and implied the idea that actually
palpation is a construct of the brain
people got a little bit of yeah no no
can't be really because you know how can
you say that the hand is such a powerful
thing you know we we feel things
uh from a philosophical perspective
people were talking about you know the
hand is an extension of the brain so you
know clinical reasoning it kind of
continues through you know we gather we
gather sensory
evidence uh by listening by observing

the patient by
arguably using most of the senses
including smell because i don't think we
taste our patients but smell for sure
you know they are there cases where a
strong smell equals pathology so we get
that information we've been bombarded
with that information
and we try to make some sense of it and
you know it's it's it's down to the
clinical reasoning
so i think that fits nicely now with
with new ideas of you know along the
lines of active inference because
it's about you know active inference
claims about how we construct our own
sensorian you know how we
sort of
integrate information change things
proactively interact and so on exactly
what we do when
we assess a patient we touch we feel we
get a reaction from the patient either
you know that hurts or i don't like you
being touched or actually feels very
nice
so the number of things that straight
away are going to happen that
immediately one can argue changes the
outcome so it's no longer
you know what i felt now
the second guy is going to feel exactly
the same thing i don't think that is
possible unless of course if you've got
uh there's a there's a fused vertebra
between let's say c3 and c4 you know yes
we should feel the same thing because
objectively there is

there is a problem there
but otherwise you know what we tend to
find how we interact uh changes the
whole kind of environment and that i
think you know active inference um
it's kind of a nice framework to start
you know exploring things and uh rather
than just being on the on the negative
side is actually to try
and go a step further and try to
understand
you know
some of the dilemmas from a different
perspective from a different angle
without saying you know this justifies
now everything we do
far from that but actually try to
understand some more that
some of the difficulties some of the
tensions
yeah
may i
please right
so
definitely agree with what george uh
just said and what i would like to add
to the discussion table is that
um
actually we started also from um an
additional point there was the fact that
two additional points so the first one
is that apparently all manual all manual
therapies has got a an outcome
so it's got a positive outcomes for the
patients so that can be an osteoporo
there could be a physiotherapist it
could be chiropractor that could be
any a massage therapist so apparently

the uniqueness of each kind of
manual therapist

is a kind of flattened by the the
relationship that the that manual
therapist can establish with the patient
so giving therefore the importance of
the the type of relationship into you
with the with the with the patient they
might also play a big role in the
cleaning positive clinical outcome
effects

and the saturn is also related to the
fact that the
um based on quantitative quantitative
research

the
elements that actually pop up is that
some in some cases and more than some
cases the the specific application of
osteopathic treatment might also be
effective and with the clinical
effectiveness in it across different age
groups so from preterms to the elderly
to the

uh to olders and so on that actually
opens a very good question in regards to
the real effectiveness because actually
we know that from science that
that that it is not possible that we are
going to have always positive effects
otherwise we are going into publication
bias so we are going to publish only
the papers that actually are um are
positive towards that that kind of
um the kind of treatment so far so and
indeed in this way um
there might be therefore um the the need
for a paradigm shift

uh towards the as george said and
mentioned from the biomechanical way and
then
modified by the um
their bps so the
the bio psychological
um
the bio psychology the biopsychological
model into something that actually might
really take into account the environment
because in the indian
and these is also important because the
evolution of the free energy principle
and all the
the elements within
uh this content these contexts actually
take into account also elements then my
from a computational uh point of view
might take into account the relationship
between
uh people for example a mark of blankets
and the all the elements actually are in
contact between
the practitioner and any manual
practitioner in this case and the
patient
for for the sake of uh the the fact that
actually we are touching patients and
therefore it's a super powerful channels
that also with or without words might
also vehiculate and drive information
towards
a specific hand
now the other elements i would like to
introduce uh is the fact that sometimes
happened that we
consider the reliability a inter
reliability between the manual therapies

perspective some
a big problem but this problem is
actually exactly the same for
cardiologists for uh people in the uh
doing ultrasound
why is that because actually it's
the uh and was a paper a beautiful paper
publishing on bmj probably 10 years ago
talking about the the kind of
statistical measure that actually has
been used for reliability studies and
mainly is the decay coin scale
that is one an intrinsic problem with
coins k because actually it doesn't take
into account what is the reality of the
measurement of the
uh the radiologists or the radiologists
are measuring the ultrasound ultrasound
specialist manual therapies and so on so
it's been implied therefore that that
time that kind of the reliability it's
one of the
most important measurement for
reproducing and for
generalizing the the uh what actually
the treatment is is doing and with this
framework actually we are trying to get
out from this perspective because within
the the context of a therapeutic
alliance and when actually we take into
account where we where we look after
patients it's not just the fact that
actually we can uh reproduce the
treatment because the treatment is just
one element of the plethora of the the
broad elements of the of the of the uh
of the instruments and tools
that the

within the the session we are applying
so it's just a reducing like a funnel
the fact that actually we need to
reproduce the exactly that that kind of
uh

dysfunction or problems in the in the
column but the problem in the column
might be also affected the way in which
we establish a relationship with the
patient so and it goes and back in the
days the bmj paper was

just uh um a brilliant uh way of looking
at the one of these main effects uh from
a different perspective

thank you awesome zoe

yeah so to add on to uh what george and
francesco said this paper mainly focuses
on more private practice than um sort of
a hospital setting obviously it can it's
a framework so it can be applied but
generally we're talking about more
dyadic or triadic relationships within
clinic um

as as the setting

um and we've primarily focused on manual
therapy but it the theories of active
influence and everything can be applied
to others like psychotherapy and things
like that as well

um just to point that out but yes like
francesco and george said we're mainly
using touch as

sort of the vehicle but there's a lot
more that's going on

sort of with the way that we communicate
and the different aspects of
like the biopsychosocial model which now
i'm sort of expanding upon but there's a

lot more to it than just the vehicle of
touch to make therapeutic alliance um i
think the therapeutic alliance is
interesting there's a lot of research
that's been done for a very long time on
the therapeutic alliance itself what it
is and why it's important but there's
not actually
that many theories to the sort of the
neuroscience or things like active
influence behind
why it's important all the research will
tell you that you can get better
clinical results
and better relationships better trust
with the patients more adherence lots of
different positive effects of the
therapeutic alliance but not always
um sort of explaining how the
therapeutic alliance works
um or how to best achieve it that's kind
of not always
um expressed so this is kind of a way
that we've tried to quantify
how to
affect um to gather all the different um
parts of the that go on to create the
therapeutic alliance instead of one
framework

[Music]

awesome a lot there ian do you have any
thing to add or i'll ask you a question
um yes thank you well
lots of tantalizing stuff there and you
know what i really love is the
openness of all of this
question asking um
several sort of things from the paper

that or topics from the paper i'd like
to explore but one thing that's coming
out to me now um is this
role of the
expert and the model of the expert that
might be leading where things are
synchronizing towards between the
therapeutic alliance and um so what i'm
sort of
came to mind as i was hearing that is
sometimes is it the
patient that's the expert or allowing
the patient to be the expert
or taking turns in being the expert so
you know it might be the the
therapist says oh you know
based on my model my you know my past
experience my priors that my the
palpation that we're doing now makes me
think maybe it's the tightness in your
traps that's causing this pain but then
actually you know the patient says
you know takes on the role of the expert
and says based on my prize actually you
know i've often had tight traps and it's
never been a problem and i'm actually
what i'm feeling is
maybe
nausea in my stomach because of all this
exam pressure i've got or
stress with my
family members and then you know then
there's another cycle of experts
swapping over and exploration so it's
more of a exploration the touch is um
helping with exploration of
the
models between

the you know the co embodiment between
the the patient and the
the practitioner
the other thing you know when you were
talking about the machines in the
hospital the radiologist or the mri
scanner um
i was thinking also about the
you know the hand as a machine and
a blind person builds up this
sensitivity to be able to read braille
um so they're palpating a piece of paper
with braille on it and maybe it says
john went to the shops and bought an
apple but the blind person would never
you know look at that pattern and think
um
that there's a person called john eating
an apple inside the paper uh but they
you know the but the symbols that
they're being created as they feel that
they'll see an image of
an apple and john and maybe it'll bring
up feelings and emotions and i just
wonder if sometimes you know it's
confusing
the sensory input with the
[Music]
feelings and how do we overcome
the feelings and the images and how do
we overcome that and you know not
confuse the moon with the finger
pointing at it
um
so yeah lots lots of lots of questions
please
look
okay

um

it's they're very good questions by the way

i think that the whole issue about taking turns i think

it's

it should be the essence of it uh one cannot argue that

it's always the expert as a practitioner if we do that i think you know we end up in a situation where

um clinical practice relies on that expert the top-down kind of approach i know best because i'm the healthcare professional healthcare provider and therefore um you know i know what's wrong with you that's that's kind of the that's away from the a model of person-centered care one expects really that

practitioner and patients or

zoe alluded to the concept of a triadic relationship

where for example in pediatric care there's a baby there and this is a conversation with the mother or the father or sometimes in geriatric care there's a conversation with the you know with

for example the daughter or the son or the older person and so on so that

taking in turns i think they are clearly uh we need to think about the person is expert in their own body and i

personally tend to ask my patients you know what in your opinion what you think is wrong with you

you know ultimately you are the expert

in your own body
and clearly if it's something kind of
new uh
you know the system becomes uh
better detecting something that you know
what's going on here
uh kind of you know something that
requires some attention but clearly when
when in in cases for example of chronic
pain the system becomes extremely good
relying on the price on the very strong
generative model too so yeah this is
pretty much what always happens when i
get stressed
you know i typically get you know start
worrying about i start getting some
butterflies in my tummy and so on and
then you know from that is just a
cascade of events that ultimately leads
to my headaches or my neck pain and so
on so i think you know clearly one
cannot talk about you know the person
the practitioner being the expert
having said that there are times where
clearly uh you need to play the role of
the expert you know i don't think you
know we need to simply just rely
on the person on the patient because
also that's why they're consulting us
you know if they if they also notice
there is a sometimes a little bit of ex
you know uh uncertainty about what you
think is going on uh this thing about
you know it's a beautiful beautiful
thing about clinical practice about
clinical uncertainty
but humans hate uncertainty anyway our
brains don't like

uncertainty anyway so you want to know
come on stop
stop talking about could be x y and z
and just tell me what's going on because
in the cases of complexity that's what
tends to generate that
that sort of vicious circle where the
guy goes to dr google then goes to the
next practitioner and you you end up
with a fascinating uh situation
sometimes a person becomes cognitively
immune to any new explanations because
hey i guess you know guess what you know
i've been talking to a lot of
practitioners nobody knows what's wrong
with me
and you know yes but you know you're the
expert so it is a a kind of something we
need to manage very uh you know
very well and and in a clever way and i
think that comes from
creating the
the environment where
listening uh
observing
a lot of what's not said but actually is
observed in terms of
the body language
is critical uh also to know when to
when to chip in and say something when
to sometimes just
shut down and let the whole thing let
them talk or let them not say anything
and
and you know and treat for example say
i think that that that's part of it uh
regarding the other stuff and we can you
know carry on with this pretty sure the

other guys will have fascinating ideas
um i think that the mental images are
part of everything you know you know you
you're talking about you know the the
blind person reading braille and so on
the early studies looking for some form
of plasticity in v1 and so on
even with congenitally blind
people they found that there was an
element of kind of possibly imagery
there's a road map so i don't think you
can dissociate yourself from the images
you create when you uh
when you're navigating the world when
you're interacting with the world and a
very strong kind of uh prior to what
we're going to
you know put my hand but i accept then
of course that that that prior is going
to bias the you know the incoming signal
and it's going to match some obvious yes
of course this is what it is so images
are done i think we just need to cope
with that
well what i cannot do what just george
said uh is that uh when observing
someone in the i mean in the clinical
practice actually
the one element might be to start
understanding the generative models
so the and
understanding the generative models
means that in addition to what george
said also how
for example the body reacts to
specific
uh stimuli to specific stimuli so also
testing in the clinical practice the

generative model might also give us the idea on how the body might react from different stimuli to uh in in for example um combining different senses so they might be so talking how the talking about for example the pain might change the generation generative models in the body for example so uh are there are some parts getting tighter or more real what happened or for example talking about or touching in specific areas how or moving the the the body in a specific way how those generative models might change so and i think it's important in the in the role play that we that the patient and the the practitioner are uh are establishing during the therapeutic alliance and i think and we discussed that also with george then the the right balance between the input the giving the um this role to the patient and to the practitioners also it's important for establishing a good therapeutic therapeutic alliance why is that because actually the patient the patient as mentioned by zoe in a way uh needs trust and trust may be also uh um might be also applied bi-directionally so in the way that actually the patient coming into our clinical practice actually they are just

giving us trust so they
by default saying okay i'm coming to
you because i trust you and because i
think you can ask me so that it's a way
of uh sharing trust but on the other
side the clinician needs also to trust
the patient about what he or she said or
what he or she uh shared and in these
kind of uh dichotomous uh role play i
think it's important the the fact that
the we need to uh understand and improve
our ability of uh taking turns
uh and from that of obviously creating
the right the right environment
on the other on the other um question so
uh the braille yes uh mental images are
always there but also mental images are
not just visual because actually they
can
during the the secondary and and and
third
um
i mean the the uh the in the evolution
of the of the sensations and the
feelings uh actually are not just
compartmentalizing to one um one image
so the visual part because actually then
are also um
characterized by other
other elements that actually comes from
experience but also comes from our
expectations so what we expect about
those mental images
um you know in addition to that there is
a difference between just you just to
pick up a very um um
a very interesting point uh
raised by ian that the uh one of the

last paper from
carvalho carvalho and um damascus
published in 2000
at the end of 2021 actually they where
they proposed the
interoceptive nervous system and not just
the interception as part of the
uh the the different uh aspects of uh
the the perception
when actually they proposed that they
also said that when actually we are
gonna have the extra receptive map and
interoceptive mapping within the brain it
actually comes from from the senses from
external senses but also from internal
sensitivity interception
after the process of integrating them
into
the the brain actually then and driving
two things the first one is feelings and
the other one the other one is drives so
the drives are is the elements that
actually are creating the substances for
creating their four feelings so are the
main elements that create therefore
the the the consciousness of what we are
feeling so far and and then uh in
addition to that when actually they are
creating those elements that actually go
back into
uh
and the the very nice special issue on
interception published on new on
on neuroscience as uh always last year
where actually these concepts this model
of intercepting model was then
modified into the homeostatic
inter homeostatic model

where we get into account the fact that actually we have the interception is creating therefore the object of what we are feeling and then there is the subject that is the brain that actually are creating a perspective of what is the interception and then there is the the the final part the is the mental imaging of our interest on intercepting condition and therefore everything uh or therefore everything so the mental images might be also checked with the real what actually is coming from the tissue and this model is called model of forecasting so creating therefore the elements between these two uh ideals so the prior with the like and the likelihood just to talk about something in the uh from a fep perspective so in what creating what what is usually called the the posterior so i think that the mental images from one side from the brain from the brain side and the braille side are then checked with something that is the perception of the the blind the blind man that actually is touching the the uh the bar with the different spots and then they are forecasting the the the two elements in order to create a mental image yesterday is a kind of a rainbow so with different colors with different elements inside in order to understand what is behind that that the story that he or she is reading

awesome

again a lot to kind of pick up on but just the way that the expectations play into what is palpated even just at a primary sensory level like anyone might be able to run their finger and detect a bump

but then one level of expertise is that integration of well where should the bumps be

or what does this mean in this person's clinical trajectory and so

it isn't just um the information flowing into the person doing the perceiving it's this active palpation in the integration of decades of the person's experience

in the clinical setting

and then the person's experience within that body and then this externalized knowledge resource that we

call like empirical grounding of science and evidence-based medicine

and then all of that like comes together in the focus of the therapeutic alliance and in the clinical setting that's like the run time where all these features intersect

and

it's just very interesting how we're seeing a lot of these threads like you you use the the bayesian terminology like the prior it's the expectations and then there's like the observations and then those are updated to form a posterior connecting that kind of quantitative approach to thinking about perception cognition and then even

action selection with the intermodal
it's like well
even if only that to combine the past
present and future
with multiple
sensations and types of sensations and
cognitive features like memory and
preference
that would be excellent for active to do
and so that um
is a really clear how it's presented in
the paper in here like what active
inference is doing
in in this um
focused setting of that
dyadic
initially touched grounded
um
relationship but then yeah
maybe there's another machine involved
maybe somebody is using an x-ray scanner
or they're using a temperature by a
thermometer that just becomes another
observation
another
mere active inference or adaptive active
inference entity in this multi-attic
relationship
so really interesting
i think if i can
add
one
one interesting thing to to the debate
is
in recent times we've been been
observing quite a lot of debate around
you know what's what's the value of
palpation you know why why bother

because you know palpation is typically
does have that connotation of biomedical
kind of approach you know looking for
something that is dysfunctional some
pathology and so on i think
one important thing is you know from
this perspective i think would be
important to take into account is
palpation still matters even if it's
just as a way of communicating as a way
of reassuring
as a way of actually contributing to
to the development of that therapeutic
niche so it's not just uh
it is this sort of palpation not being a
one-way stream where i kind of look for
what's wrong and i know what's wrong but
actually the dialogue and
that dialogue can involve of course
finding you know something
that is
you know not
not good you know potentially even
pathological to something that actually
you know that that that exploration
actually feels nice and comforts the
patient and so on so
it is important not just sort of be on a
on a kind of very um fundamentalist
approach is either one thing or another
but actually try to kind of you know sit
in the middle and try to observe the
two sides and even the tensions between
the two sides you know biomedically
based or
more so to exploratory more sort of
niche construction more sort of active
inference in nature

thank you sorry

[Music]

um yes so i think back to the expertise

i just wanted to add in about um

we brought up the concept of a bird song

and i think that's kind of how

i see it as expertise i would never tell

a patient

personally that they are not the

expertise in their pain

i don't know what they're feeling in and

their priors and different things that

are going on to make them their pain i

would always say that my patient is an

expert in what they're feeling in their

pain whatever

that may mean to them um i would say

that they are the expert in that but i

think it's it's a bird song it's a

taking here it's it's it's

you're sort of adapting what you think

yes often as a practitioner you're seen

as more of the

expert or your precision of maybe the

physiological side of the condition is

more precise than

what your patient is they've obviously

not 100 certain what's going on

otherwise maybe they wouldn't be coming

but i think it's definitely

you adapt what you think based on what

your patient is feeling and what they

say if you may think one certain thing

and then they might tell you that

actually you know they did this at the

weekend or

they felt this or this happened before

and you kind of constantly change

and take it you adapt your idea a little
bit you adapt their idea a little bit
the whole diagnostic process is
generally not
cut and dry i mean some situations are
but generally speaking we do not know
100 what is going on um in in a clinical
space without machines or without sort
of more in-depth knowledge we're going
based on what we think um in our opinion
so i think it is definitely
there's no strict hierarchy in terms of
you are just the practitioner is just
the expert yeah we may be the expert on
one aspect of the condition but not
necessarily the aspect on
the whole big picture once you take into
account all the other things that
contribute to their pain
it's not just
a
physiological response
often as the whole
to be in sort of an overload of
aliastasis you've got to have
a lot more than just
pain there's a reason why behind there's
a certain few things that probably have
gone on for them to be in that situation
so i think it's it's too simple to say
that that you're just the expert because
you've done a degree and you're meant to
know
sort of the biomechanics or the
physiology behind a certain condition
um as for
what everyone um
was saying after towards the braille and

feeling

touch is there's a lot of research to

show that touch is intraceptive it

affects the um and activates the insula

which

gives you sort of that connection

between what you're feeling um your what

sensory stimulus you're detecting and

the feeling that you contribute to that

so

that's why

sometimes we have a touch and we feel

it's pleasant

um it's that effect of the insular and

how it is processed and i think that's

obviously dependent on

the individual different people will

have different things that will

make certain things feel more pleasant

or more painful depending on

lots of different aspects and so

i think we touched upon sort of touches

a way of a good way of non

nonverbal communication of expressing

not just examining the condition but

also

showing that you understand through sort

of empathy and things like that so

you

you can use touch to show what you're

feeling without saying i think this is

something that

is very evident with the pandemic i mean

there's a lot of people that went

through a lot of hard times and the one

thing they wanted to do of everything

was just give

the person they loved a hug or

you know they people missed funerals and
weddings and
lots of different sort of events
significant events in other people's
lives and all people wanted to do was
just
you know
hug the person if they just lost someone
so i think
touch has a way of
communicating with people on a different
level than not just
sort of the
purely clinical level

[Music]

thank you and um yeah just followed on
from what zoe said it reminded me daniel
of um

the example you took from the very
beginning of the new active inference
manual of um

you know

putting your finger on your thigh and
saying

is this rough or smooth and you have to
kind of uh

you have to move your hand to tell if
it's rough or smooth and then i'm

thinking about you know animals preening
themselves

so preening each other or grooming each
other and you know

proposing here from an evolutionary
point of view

we've um been able we've found this way
of distinguishing between

um you know a nice smooth something
smooth and soft

that's supportive and bonding and packed
like animals building a pack
together versus something that's more
like a fight or a clash or
um conflict
and um
you know that in the paper you you
talked about um
we're coming arriving at this sort of
insight that we're all identical i think
it was or very similar
so that brings me back to
something like safety and
um
a likeness and not reject the role of
touch in
um
feeling part of
a pack or a tribe versus
um being rejected or outcast or fought
away
so yeah you know that role of touch in
reassuring that everything's okay we are
similar um
the other thing what uh um francesco's
left but he talked about the
understanding the generative model and
as he was talking about that i was
thinking about
i don't know if anyone's um
come across something called hakomi
it's a form of group
but can be done group or individual work
where a bit like somatic experiencing
where the participants
um
try and understand in a sort of
reflective listening each of what's

going on inside each other so try and
explore each other's
interceptive world plus also what might
be going on in their outside life
and i remember hearing about performing
little experiments so the person who
takes on the role of the expert kind of
says okay

let's do a little experiment here what
happens if

you know what happens if you
say this or do this or take this action
what does that do to your
your experience

and you know i think that's kind of what
happens anyway in therapy you know
physiotherapists will give people a list
of exercises and say you know do this
experiment and hopefully i i hope that
you'll get better um but i'm just
wondering with this sort of paradigm
shift of

uh

the physiotherapist just being the
expert and prescribing something that's
definitely gonna work it's more framed
as okay you know i'd like to encourage
the patient to be more experimental um
rather than just be prescribed something
whilst appreciating what you said you're
get your uh george about um

you know they they actually come wanting
their hope to be guided in a certain way
so you know we need we need to have a
certain level of confidence and guiding
but not necessarily
being too prescriptive
um

now the the thing about you know the
generative modellings and how we can
change things

uh

so for example you know these two
two sides are the same equation i would
say uh

one is when you for example in patients
with um

as you phobia fear of movements when you
ask them you know for example you know
can you perform this kind of action
typically for low back pain most people
are scared of

bending forward going into flexion
because it's typically associated that's
how my disc is gonna pop and that's how
you know i ended up in this situation
and so on so uh

you know putting in you know
perceiving a little bit of that of that
generative model which could be one of
you know

moving forward going into flexion equals
pain

uh you sometimes just by telling okay
let's try to do this movement and
before the patient actually starts doing
the movement you find straight away an
immediate muscle guarding you know
is a sort of oh yeah you know i can't do
that that's that's hurting straight away
and and in fact you find sometimes that
muscle guarding so you know where does
palpation actually come into the
equation you know rather than being
diagnostic you can actually
also look at you know what's going on

you know so to try to align our mental states with them and predict what's probably going on going on now from an active inference perspective you could argue that if they believe them of the the priors is very strong that flexion equals spine it's likely that kind of a minimal a minimal sensory input is likely to create that muscle spasm and therefore yes i knew i would do this and it would be in pain okay so that's what sometimes we we observe and actually if you start thinking about some of the some of the clinical signs and symptoms you find with patients with persistent pain for example in people you know either catastrophization or you know is a phobia and actually putting through this perspective you can argue that you know what's probably likely to be very strong is a generative model sometimes with movement or this type of action equals damage therefore i have more pain how can we short-circuit the whole thing how can we start changing a little bit of that generative model sort of in a sense creating a prediction error that is so great that actually surprises that the system um so for example you know one could use for example breathing or focusing on some interoceptive sensations that's that's been used by other practitioners for example

in um in many you know musculoskeletal
care the concept of cognitive functional
uh

you know um therapy is based a little
bit on that for example using breathing
asking the patient to sort of focus a
little bit on your breathing take a deep
breath in relax and not try to move the
movement again and you find that
sometimes you know there is an
improvement on a few ranges you know and
you know degrees of range of motion
but arguably if you think from this
perspective actually you started
changing the model say oh patient
sometimes looks at you so
that actually feels a little bit easier
how come what have you done you didn't
do anything you started sort of uh you
created this you know surprised the
system okay

and you surprise the system in a very
positive way which is yeah you know that
that's great you know um

we explore this on the on the paper that
is should be out very very soon which is
try to understand actually how can we
generate in a change degenerative model
with guided touch for example creating
conditions where

one could say okay let's focus on uh you
know let's focus on the sensations let's
focus okay you know this feels quite
tight and actually what i'm doing here
i'm releasing the fascia and so on okay
so you can use that if you think the
patient is actually
is a suitable candidate for that you

know you're not actually implanting
another generative model there's
something wrong with the structure okay
that could help but for example if the
patient gives too much attention to to
that to those sort of
sensations
then you know it's probably wrong to get
them to focus on those sensations
so because actually just perpetuating a
little bit of that you know belief that
prior that actually okay it's tight yeah
i'm worried about this and so on so for
example we proposed these things coming
from kind of a
traditional sort of um
meditation buddhist
practices in in um
you know
that's for example you can short circuit
the whole thing by for example start
talking about something else
which we talked about flattening the
sensory states okay so imagine that you
know you're palpating you're doing a
technique and so on and suddenly and you
feel like the patient is giving too much
attention to that actually
there's a sort of failure of sensory
attenuation if you want to use in all
the jargon here but
if you if you then suddenly talk about
oh yeah you know what i've got a new dog
i was like where does the new dog come
from okay
or you know if you think that's uh you
know they love football for example and
their team won last night they say oh i

watched the the match last night oh it
was great wasn't it
so that takes the attention away from
that and actually kind of
you know maybe enables you to start
changing something that so it's a
combination of clearly hands-on stuff
where you
cleverly can introduce the language
to start
start surprising the system start
changing a little bit of that generative
model so
i think that's okay it's uh you know uh
colleagues in the uk um
have developed this osteomap approach
those jobs in mindfulness and uh based
approach also based on third wave
cognitive behavioral therapy and
actually uses the similar thing which is
you know you're treating but you sort of
talking you try to guide sometimes guide
the patient say how does this feel
what's what's going on here you know why
why are you protecting and so on so you
start exploring this so that could work
really well arguably in a person that
interceptively
able to explore that
if they're not
interceptively prepared so for example
switching you know giving too much
attention to what's going on insights
for some people makes them even more
anxious
so there's no point in saying yeah let's
mentalize interception because you know
that's going to be great for everything

no it's not actually now i feel like a
rumble in my tummy could be something
going on really isn't it oh you know
what what about if there's a pathology i
just consulted with dr google he told me
about this stuff so for some people
that's clearly a no-go zone for other
people yes you know we can explore that
so again if you think through this lens
uh i think that there's a lot of uh
you know i think one can argue you can
have a lot of fun in clinical practice
by exploring things
rather than limiting yourself to a very
rigid sort of boxed
approach
and sometimes you you get a patient
who's you know they get up from bad from
from the you know from the plant and say
oh actually
feels a little bit better you know
actually and it's it's painful but it's
not as painful so
arguably you started changing something
already and that's where sometimes okay
is it uh is it the sort of expectations
is
a placebo phenomena it is what it is you
know you created all the conditions you
put everything together it actually
created a package of care
that arguably can start changing things
and for me and when a patient says it
actually feels feels a bit easier
it's kind of okay that's good stuff you
know it's it's not the result of my
technique is sort of whatever happened
in this setting here

there is you know they started the road
you know their journey in the road to to
recovery and that's pretty good
and then you you've got you know if you
ask the patient sometimes after a few
sessions how do you feel today you know
i feel better i feel much better how's
your pain
now on the visual analog scale the pen
could be
not different from last time or even
this is the first session they may have
started with a
you know six or seven or eight or eight
out of ten and at this stage are on a
six out of ten for example
so clinically there's not a meaningful
change anyway but they may actually say
i feel much better what does that mean
you know feeling much better is yes you
know it's the interceptive stuff it's
about you know i
i started kind of taking you know
i started changing something so
there is
i think
stuff to play with that combined clearly
not just hands-on but the whole kind of
uh you know creating a a package a
multi-sensory um let's say experience as
well
awesome i i'd like to make sure zoe
please
um just to follow up on what uh george
said um
just from the the last bit and then i'll
track back to
um another point but i think

it doesn't really matter sort of how you
do it whether you try and shift their
attention away from it by talking about
something else whether you shift it away
from

using breathing or trying to approach
the

way in a different way you're basically
trying to shift the attention away from
what if they especially if they have
chronic pain is an overly sort of
precise maladaptive belief i mean
it's just trying to get them to kind of
step back a little bit from the prior
that maybe or the feeling that they have
predicted that they're going to feel so
if they uh like george said bend
forwards is painful if you get them to
step back a little bit from that whether
you do that

um

through

talking or

sort of focusing on your breath through
mindfulness

you

want to shift it back to the the current
what they're feeling rather than what
they think that they're about to feel um
so you can

do that in many different ways but yeah
definitely to try and focus away from it
um i think it's you either kind of go
down that path and shift the attention
away to kind of then try to refocus or
you kind of use touch is what we've um
done in the paper to
try and develop new associations or to

reframe or to overwrite the the prior beliefs that maybe aren't quite working for that situation um to try and hope that by reducing and rewriting it by doing those exercises maybe to try and get them to do something that they they fear they think if they bend forwards because when they hurt their back that's they were bending forwards to pick something up from the floor and that was painful so now they're absolutely terrified that if they bend forwards that's going to happen or then dad sort of blew their disk if they when they were bending forwards to lift something that sort of prior belief we just want to try to make the opportunity often through touch or by doing an exercise and giving that sort of feedback that actually yeah you're you're bending forward but actually it's not that tight um and to try to inhibit that prior belief that they've they've got about expecting the pain and tried to reduce the weighting um that they've put on it so maybe think oh well and even if you can't change it so that they stop believing that every time they bend forwards it's going to be painful even if you can try and reframe it to context specific priors so say like well if you bend forwards in this way then it's okay you might not change the fact that they still think that bending forwards is um

risky that

you know they might not be they might be
cognitively immune to to changing that
completely but you might be able to
through touch and through sort of
encouraging them

um to develop the prior for that context
so

if you

you know bend forwards with this posture
or this way or that that you can um
you can update it and change it so they
think okay well maybe actually in that
situation it's okay

um

this brings us back to um

they did a study with little kids about
um with

one that was slightly younger one that
was um a bunch that was slightly older
on balance and they gave the younger
kids thought that balance um

was just

sort of gravity so um things balanced
because they had gravity working upon it
whereas the older kids thought that it
wasn't that simple and that if the
object was weighted more on one side
then obviously even if it was gravity
pulling it down they'd tilt

and so there was two different theories
of how balance works

and they kind of the um

the investigators gave the kids the
other

one so all the ones that thought it was
just plain gravity were introduced to
the concept um from the other ones and

vice versa and they explained these concepts and then they showed them the block which they'd kind of used to say like look this is the block this is how it balances this is your theory and then they said well do you want to play and almost and every single one took the block tried out the other theory of how it balanced and then after sort of 10 minutes of play they asked them what they thought again and some of them changed their their minds and they used that sort of tactile sort of learning and touch to to update what they thought about their theory of balance and then to change and adapt slightly of what they thought some didn't change it completely some did but they all wanted to touch and explore to give them the opportunity to sort of understand that prior and to build up um a new belief of what maybe they thought was going on so i think we do that a lot in clinic um [Music] sorry just lost so look you can move further now or it doesn't feel so tight or that knot that you said you had doesn't feel as hard things like that um and back to the comment about how we relied on others i

think it was um yes i'm working together
um the hunter gatherer thing um
they i think it's important to note that
we're

we're built to work together i mean we
haven't got a choice um we might when
we're sort of old enough to go hunt for
deer and boar

and things like that but when we're
first born we we need the touch of our
mother or a kegger of some sort to get
to get our allostatic needs we cannot
survive without them um if you leave a
baby on its own

that
they probably won't survive on their own
they need the mom or the person to to
pick them up to feed them to regulate
their body temperature

to to do everything they they don't have
a choice and even before that um
neonatally

you're constantly as a fetus being
bombarded with different information
from your your mom and we touched upon
this in the um

the paper you need

they they they constantly know what
their mom is feeling and so of course
once they're born they still want to
coordinate with their mom because
that's what they've been used to for
nine months of as they start to develop
and start to feel touch in different
areas and you know they start to get
nerves they want to explore and see what
happens so

yeah we're inherently need to rely on

others at some point and while we get older it's not as continuous and we only rely on others maybe when we don't know what to do um

then then we will it's not so like i'm not gonna relate to my caregiver now i know what my heart rate should be i know what's normal for me and that's different but when you're younger you don't have a choice and so

and as you can't speak touch is is the first point of communication so yeah i think i'd touch on most of the points that were mentioned earlier awesome

i wanted to bring a few active inference points in and and loop it in and then ask a great question in the chat so we've been talking a lot about the generative model and so the generative model is a statistical model that includes all these features we've been describing like perception cognitive variables and action selection and also it integrates this bayesian idea of the prior and the updates and the posterior so we talked earlier about how the practitioner is engaging in these informative perturbations and experiments of the patient's generative model

and those perturbations can be seen as action selection

that are informative by the generative model of the practitioner so in the therapeutic alliance in the niche of

that clinic there's like multiple
generative models that are engaging with
each other and so that's a setting that
active inference has been used to
study
and to connect that back to the bird
song this is just it's it's such a
fascinating um avenue of research
that i think it's worth going one level
into how the bird song plays out so
we're we talked about the conversation
and about how the patient has one type
of experience and expertise
in their own embodiment and then the
clinician has another type of expertise
in their education and also in their
embodiment and where that comes together
is through touch so in the bird song
example there's this expected bird song
we can think of it as like the
stereotypical or the archetypal bird
song that the birds are expecting to
hear like if you're listening to a song
that you know you're expecting it to be
playing out a certain way
now
why
doesn't one bird just sing
and then the other one is hearing what
it expects to hear
so it never takes a turn and so in other
words like why what the equivalent in
the in the clinical setting would be
like the expert clinician could just be
speaking from their expertise and just
giving a one-hour lecture on a condition
or you could have a situation with a
one-hour lecture the other direction

with just the person sharing for one
hour so how do we move from that like
one directional
um
non-turn taking which still could be
absolutely expertise on either side but
there's this like emergent joint
expertise when there's a conversation so
how does active inference model that
turn taking
and
perhaps not the only way that this could
be modeled but how it is modeled in the
bird song papers of Friston and Frith
is that action selection
requires a transient suppression
of precision on action selection
so in other words if i'm seated right
now and i want to engage in standing up
there has to be a transient suppression
of precision like an openness around
what am i doing because if my confident
belief is i am sitting down
then engaging the counterfactual i could
stand up will never be enacted because i
have a high precision belief that it's
simply not the case and in fact they've
connected that to various
movement disorders where there's
challenges in initiating or challenges
and terminating behavior and so what's
happening with the turn-taking birds
is that one bird is singing and so it's
singing from the song book so to speak
the expected song
but it's continued engagement in singing
is accompanied by this increasing
reduction in precision

around what it is doing it's kind of
like someone going on a rant and there's
like they're actually losing precision
they're losing the thread as the rant
continues
as a function of them suppressing
accuracy because they have to engage in
action
and then at some point that elicits
that acting entity
to seize action
and to increase its its precision
through listening through active
listening and so that is how the birds
take turns in this bird song even though
they're singing from the same sheet they
do take turns because they're engaged in
like a transient suppression of
precision on action that allows them to
act then that imprecision around action
becomes too much and then they need to
recover
that precision through listening again
so it's quite an interesting thing and
it just reminded me of like well there's
a dialogue it doesn't have to be equal
time it doesn't even have to be speaking
all the time but there has to be this
emergent dialogue
otherwise
it would just be a one-way phone call
with the download on the clinical side
or the download on the person's
experience so i hope that that these
kinds of formal models like the bird
songs the toy model but seeing that kind
of a generative model play out
in the context of the skilled

uh performance and experimentation
this might sound weird but you know why
don't you try doing this and then the
patient brings up something
unconventional or they discover
something new about their body and that
is where like healing and and openness
can occur so that was like awesome
threads
and then i want to ask the question in
the chat for for everybody it would be
great to hear their responses so
glia maximalist in the chat has asked
how can patients learn to think about
pain
in order to update their generative
model
or is this something that is possible to
do subconsciously with the assistance of
the treatment
so what are the explicit
cognitive and implicit cognitive
aspects
of updating generative models around
pain
yeah okay um yeah george then denzel now
okay
um
i think it starts with education about
pain you know what's
what is pain and why people
have that experience and clearly having
you know two
sort of situations where uh in in acute
pain
pain being kind of regarded typically as
as as a useful protective mechanism if
you just hurt yourself you

you know if you twist your ankle and
you're in pain uh and there's a minor
tissue
problem and that needs time to to heal
and so on so clearly there is pain and
people need to understand that you know
you're in pain because this happened and
this protection and should go away and
so on the the challenges is is much more
complex when we face chronic pain
because you know in those cases and a
lot of times the protection
and no longer needs to be there why is
is protecting from from what and it's
not necessarily a need to
to protect but the system becomes pretty
good that's you know it's almost like
generating that stuff and people
continue having sometimes a presentation
that is in a limping or or a particular
muscle spasm clean movement and so on i
think it's it's about sort of
starting with with understand getting
the person to understand the nature of
what's what pain is about without giving
them a lecture on the neuroscience of
pain because you know that that's
clearly does not work as well so it
needs to be something that is simple
straightforward so it starts with that
there needs to be a
you know
a reasonably well explained um
kind of framework
um
you uh i think you know what are you
talking about the bird song and so on
it's kind of you know

interesting thing about you know sensory
attenuation which
in in in the context of of chronic pain
and we argue that on the other paper on
active
osteoporosis in active inference
that probably in some cases of chronic
pain is is actually a failure of this
attention so disengaging from that
um
constant attention to to the problem how
do you achieve that
yes
again
communication touching talking or not
talking and taking turns and and and the
analogy to the bird song is is is
clearly you know it's a fascinating
thing it's clearly
what we try to to reach you know that
that synchrony between two or more
individuals in that clinical setting
when things start
kind of working well and uh you know
it's leading to somewhere so just some
thought so in
thank you
yes i
you know
agree with agree with that
that needs to be kind of simple and
understandable explanation but uh
challenging the
the perhaps previous views um i'm just
sort of reminded of um
specifically for the person who asked
this question
the work of yoni ashar and torvagner's

pain reprocessing
therapy
study is a bolder study
but it's been
converted into a book called the way out
um on chronic pain
and an app called curable so um for
people who are watching this live stream
there are some simple things to to check
out and the way out is published by
penguin and the the elements of how they
suggest to
um
self you know self
care for chronic pain are two things so
one is they call it somatic tracking
which is basically an interceptive
exercise where you with curiosity
and lightness and openness sort of um
explore the pain
and um secondly is to build evidence of
safety so to actually literally write
lists of you know right now i'm
experiencing
um
evidence that my body i'm safe in my
body for whatever reason it might be
because you know just i'm here sitting
and breathing that means i'm relatively
safe
um so those two things exploring
the
the pain in a different way and building
evidence of safety um so that's you know
i i think that's a quite nice starting
point
thank you zoe anything you would add
there

yeah sorry

i i was dying

um

yeah i think definitely as george said

trying to keep it simple is

essential if i start talking to my

patient about like the acc and how it

changes this part of the brain and it

works on this bit i think i'd lose them

within five seconds

um they're not

most patients while they're not so

interested in all the neuroscience

behind it um if you tried to to speak to

them i think it depends on the patient

how best to sort of

um

make

their pain kind of realized i think most

patients find it difficult to say what

their pain is sometimes

they can't describe what they're feeling

they they don't know the words to kind

of say which is why sort of i think

touch is also useful in this aspect

because you might be able to

try and show

someone something without actually

having to to say it

um and definitely as george said in

chronic pain it's again not so simple

people can be cognitively immune no

matter how much you try to explain it um

they might just

not want to

sort of update or to to hear it so

um

yeah i think it depends on their priors

their situation and how best you think
that you can kind of get that across
whatever way um you think might be more
amenable to them
i'll pass it back to ian this yeah yeah
thanks um
just picking up on what george and zoey
have said about you know different sub
sub categories of patients who might be
more or less open to different tools
that
can be offered to them so as you said
you know someone who's suffering from a
certain type of anxiety might be
immune to
bringing more attention to their
intracereptive sensorium whereas um
you know
that might be accessible to someone
who's cognitively immune to the can you
know the conceptual idea they might
actually
enjoy the
so i'm thinking for future research how
can we
find markers or indicators for these
different subcategories of people that
can be picked up really early so you
know a therapist can say work their way
through a flow diagram when i'm doing my
taking my history case history for this
patient they you know as i'm talking to
them
they seem you know they their own
they're showing signs of being open to
this thing but not that thing and then
you can work your way through and end up
with okay i'm going to use this i'm

going to try this tool first as the
first test and then if that doesn't work
then i try try that so you know getting
some
big studies with big n numbers that um
that try to identify these different
subcategories of
clinical indicators

that's awesome thanks zoe

um yeah so

i haven't read massively big studies but
one thing to consider that you can get
from the case history is the family um
especially if you're treating kids the
research that's come out um more
recently

on the parents priors

um they took a study looking at kids in
a rheumatological clinic

93

of their parents had pain

um had chronic had had some sort of pain

um they showed that if the parents
reported pain when they were doing
exercise the kids were more likely to
have higher pain levels to um report
more problems not just in and general
health issues not just with that but
also

things like they didn't want to do their
homework um so even more general things
were affected by sort of the way that
their parents thought they were more
likely to catastrophize that was one
um massive thing that if their parents
had experienced more pain then the kids
risk of catastrophization was
significantly more and almost that you

could predict
the amount that they would from that
same with studies
with oxytocin the parents oxytocin from
10 minutes play with their kid then men
that you could empirically predict how
much oxytocin the kid would increase
um so i think
some things like that you
can
pick up from the case history if you
know people or they're willing
to let you know
about
what might be going on previously some
patients won't want to talk about it but
some
you might be able to get from if there's
more than one of them in it or something
like that that's one indicator now of
what you know
if you they come in and someone else or
they've said something about
um a belief that someone else has had it
might be harder to kind of
change
it like why are you more um
of an expert than their
parent or whoever had said it so i think
that's
there's one way that they're starting to
kind of pick up that you could easily
implement into your case history you
might already be asking about that you
just didn't realize the significance
of it you knew it was significant but
not the extent that we're now sort of
starting to realize

yes

very interesting of course we're just

exploring like different

avenues but in

and it's not my area as only an ant

doctor

but

whether somebody uses

thinking or feeling type words

or refers to their own direct experience

versus the experience of somebody who

they trust versus information they

received from a search engine

those are all cues

maybe that is what then

the clinical decision making pivots

towards

or maybe it's acute that other

areas in that cognitive space would be

informative to explore like if somebody

spends 10 sentences talking about what

they received from their body

you could ask if you've asked anyone

else or what did you look at when you

researched it if somebody says 10 things

about just what they researched maybe

how does that reflect with your own

experience like just to kind of even the

score and and sample across different

regions of this space

then that could potentially you know

help

give a coherent and holistic

perspective which

is kind of implicit in a lot of the

discussions that that all of you are

bringing here so ian

yeah all really good

um

suggestions there and

zoe as you were talking i was thinking

about um

i've heard sort of similar studies with

uh parents with digestive problems and

you know children with digestive

problems but then it got me thinking

about intergenerational trauma and dna

methylation and you know epigenetics

um

and whether they're you know this is not

necessarily within our scope immediately

but longer term are there any

dna screen methylation screen patterns

that can be used to see if someone might

be more or less um

open to

different

interventions

interesting

note there like and these these

cognitive approaches it's a it's a

complement to understanding the

correlation between the rheumatological

outcomes of the parents and the children

like one

possible aspect is there's a

predisposition

carried through

the dna sequence carried through an

epigenomic modification carried through

a shared and cultured niche a shared way

of expecting things like

those may come

differently into play

in the n equals one

that is the therapeutic alliance that's

assembling around this person who is
having this experience and so that is
kind of like what breaks through
the the large n
and just presents us very plainly and
starkly with this one person's
experience and there isn't going to be a
large study with that person because
nobody will ever replicate
one or two or five of those aspects so
that's like where the the um
the large and the small end and sort of
a top-down and a bottom-up approach to
these topics
hits the the compromise um george
um
yeah that would be fascinating to try to
see if people are more
predisposed to respond to some forms of
therapy and or not now
interesting thing you know we we've done
uh several studies in you know in
neonathology uh
with preterm babies and we found that
osteopathic care seemed to produce well
was effective in reducing length or stay
in hospital we tested also the
hypothesis comparing
in a static touch with a form of dynamic
effective touch we saw that um
you know there is a there is a role in
modulating the you know reducing arousal
from from an autonomic perspective it
seems to you know babies respond to to
death
and one could argue that actually
they are you know that the priors there
aren't

great yet because you know they're responding to to that to that contact we also know that kangaroo care is an effective evidence-based approach in pre-term for preterm so skin to skin contact with with with parents so this stuff seems you know again plays you know suggests that touch seems to play an important role it would be good and we tried to do that to look at for example the babies that received that hands-on intervention uh to follow them for a number of years and do prospective study and actually see what happened uh you know in terms of cognitive and you know uh you know and and general sort of development this but then you know if we if we start thinking about okay yeah it's okay the babies are sort of not uh don't know really how to respond then there are some some interesting ideas being developed in particular with people like anna chanukah looking at uh inutro being the first prior in fact okay so so and if you look at in some cultures mothers touch their belly and sort of interact start interacting with with the baby very early on uh so actually arguably those babies already starting being touched and they're sort of primed to probably responding a bit in a more

positive way to
tactile kind of stimulation
and that is also uh research start
talking about for example in the case of
of of twins
they start touching each other around so to
week 12 a week 20 or so so again there's
a form of you know
tactile contact
stimulation
uh other you know other authors for
example Katrina Fortune talks about and
actually the first prior is actually the
mother for example so again goes to the
idea that actually either through um
you know prior to
one being born the the sort of that
interaction with the mother creates that
price so likely
you know it creates a predisposition
perhaps to to
to the sort of a good response to a more
manual type of of intervention so
you know the fascinating things can be
explored of course you know always have
you know from a negative perspective
sometimes in randomized controlled
trials people say yeah but you know
those guys kind of you know was good
outcome because those were good
responders to treatment anyway
but again what does being a good
responder to a manual type of
intervention actually means you know the
variables are such you know so many that
it actually
um you know kind of
the scope is huge so

one one could explore but actually there
are some serious questions in you know
in the whole thing for sure
thank you
zoe and then we'll have kind of a
closing thoughts as we finish the dot
one
yeah i think just to go off what george
said they've shown in
um neonates as well um
that as
basically as soon as an area of the body
starts to get um
any sort of nerve innovation at all the
fetus will start to touch that area so
when they start to get the face which is
often one of the first that's why you
see a lot on ultrasound's baby touching
their faces or their hands or things as
soon as they get innovation to that area
they're touching it to explore to see
what that
what is this
um that they can now feel so i think
definitely touch is um one of the first
things that
people will use to develop a prior or to
just understand what is going on in sort
of the world that is around them um
to expand upon what george said the
long term they did do one study
about the long-term effects of a
kangaroo care
incubator and full-term
um babies and they found that
um while the this was more towards the
the synchrony with the the parent but
when they

were

initially it was the

the full-time ones that didn't need any
help had the sort of strongest synchrony
the strongest sort of

um physiological

yeah

sorry i saw miss midnight here and um

then the

um skin skin contact um

was sort of next and then the incubator

was after and this was initially um one

they were they're doing it but when they
followed up on these same kids when they

were adults they found that actually the

the synchrony between them when they

were adults to their parents um was the

same so the ones that got the skin to

skin contact had the same as the

full-time ones um so it had developed

over their whole life that continuous

effect of touch builds up um so that

they were on par whereas before they

were lower down so it's not something

that's just developed

um

in those sort of critical

period where things are developing more

it is something that you know the

effects that had happened when you were

one two three um to building up when you

were 12 to 18 still have an effect when

you're not i mean there's areas of the

brain that don't start developing fully

until you're 26

so you know that's a lot of experiences

that you're going to have from different

parts until that age they're all going

to contribute to it so it's not sort of
just

the first prize then they've shown that
those prizes especially with touch are
still evident

in adults

just because they had its baby

it's another value for integrative

models we can have continuity between
the nutrients delivery

in utero and then the neonate with the

spoon coming in airplane making the

landing and then adults who have a

little bit more agency maybe in their

peri-personal space with the spoon and

landing the airplane but there's the

supply chain so it's just another part

of an extended

nutrient cycling system

and

are we going to have custom

narrow theories for each one of these

settings

and or

will we have some layers and

interoperability between these different

settings that help us like transpose

useful models and find areas of contact

those are all really important

um areas for lifelong learning like you

brought up and there's areas of the

brain that

continue after you're 26 people still

learn things and so that's brain

development too um

i just wanted to look at figure one and

or figure two and just in kind of our

closing thoughts

what is uh one of well figure one and figure two both have like a really creative and um delightful style so figure one is focusing on different functions that are discussed in the paper and in relationship to empirical evidence around different brain regions and figure two is kind of pulling out one level looking at different functions that are arising in here the dyad so maybe if each person whether figure one or figure two what's like one area that you're excited to talk about in dot 2 or one thing that you think um like really stood out to you as something that was a contribution that you weren't seeing in other work or in other ways of representing these topics so maybe ian first um and then we'll hear from the authors but anything from figure one or figure two that would be exciting to explore more thanks daniel what stands out for me today is the two monkeys doing a one-handed handstand with a joyful look on their face so this idea of cooperative communication and that to me today that speaks to this idea of you know they neither of them look like they're an expert they look like they're playing together um which is a nice approach uh to think of in the therapeutic alliance um not always

possible to be playful but you know when
possible
can we be like the monkeys um but then
and moving on
i'd like to explore maybe this idea of
um
arriving at the realization that we're
similar or identical so again those
monkeys look pretty identical but
what does that mean i'd like to hear
from the authors about about that
and then
maybe next time or exploring this idea
of
when the generative models or the
allostatic
regulation the set points start to drift
and that you know i'm thinking more
broadly in our society our health
global health problems
are
are they drifting of our allostatic set
points around food production and what
healthy food actually means and how it's
how that's
aeroplane coming into the mouth you know
the some
expert model has um told us that certain
foods should be
what we eat but the you know the health
indicators um
suggest that it's not that model may not
be the best model and how do we as
therapists are open to this active
inference model kind of keep that
possibility of drifting this model
drifting and how do we how do we decide
what the set point should be

yeah from the from the airplane landing
the spoon to literally
the airplane landing with the food
we never grow up
okay figure one or figure two george
um
i have no
i don't think i have a much more to add
um
you know the the figure figure one of
course you know
it's a sort of an overview really where
uh potentially so there's a sort of
potato um
effects of touch at all different
different levels
um
yeah you know
one thing i'm quite keen actually that
thing about the two monkeys playing and
kind of communicating and so on is that
although we we
you know we have a strong bias in the
paper towards a certain neuroscience
kind of based model
that uh it is critical that we don't
turn the whole thing into a neurocentric
model okay because
ultimately if he complained too much
about biomedical model
and then kind of
pain management became
brain-centric neurocentric model kind of
you know
nothing really changes does it you know
it's a sort of still
you know uh
depending on an expert that details

tells you exactly what to do so um yeah
that's i i like you know the figure two
kind of gives us more a little bit of uh
an ability to look at the the whole kind
of process
uh where uh within that process and
again
i'm i'm quite aware of you know
criticisms that
you know this paper may also attract
from yeah you know these guys just
basically quantifying around touch is
the most important thing and so on and i
clearly i'm not on that at that level
i'm i'm clearly arguing for no no it's
not that is is an important tool
it's an important vehicle but ultimately
it's about communication in my opinion
it's about me you know what's what i uh
what i'm trying to kind of get into that
niche that bird song yeah uh getting
that biobehavioral synchrony going and i
think you know when you're getting there
then things you know can you know some
very interesting things can happen
and uh and and you know i
you know we can explore this that's uh
2.0 next time and
even
some ideas about some of the techniques
that potentially people kind of make big
arguments and people are super critical
about you know for example more cranial
sacral or cranial-based type of stuff
that it doesn't work that way but you
know one could argue that maybe the
phenomena that we observe is actually a
phenomenon of synchrony is that is the

bird song
and that enables us to get some very
interesting things going so
i just i'm i'm thinking a lot i'm
allowed i'm kind of uh
exploring through but uh
i think for me that's so figure two
clearly the other one is sort of you
know
we don't treat the brain who treat the
person
there's a person living that or in that
body
uh rather than treating bodies you know
we're not
you know don't think we should sort of
be at that level you know it's body work
you know it's not body work is you know
it's it's a different story you know
it's really it's about the person
more than the body the body tells us a
story but it can be a mark of blanket
body can be many things but
body it provides a landscape
sensory landscape an opportunity to get
in to kind of influence and so on but
okay that's it you know what about the
person lives there and what's uh that's
that's for me the fascinating bit you
know and sometimes the thing you change
minor things in their generative model
when the whole thing starts getting
better anything you know
you surprise yourself so
that's placebo yeah whatever it is but
you know that's i think that's that's
could potentially be the beauty of it
yes from kind of a body work

the cadaver materialism physicalism to
embodied work

with the richness of the situation that
that's the map and the territory and
very interesting and the last word for
the first author

yeah so i think um both the figures um
i think figure one

when

i was sort of

sketching out what i sort of wanted to
do for i did again sort of go with um
from what george said it's not yes it's
neurocentric in terms of it's labeled
based on the areas of the brain but if
you think more sort of broadly about it
the

i mean it covers a wide range of things
you have the the two monkeys having fun
to the mom um

holding her baby to

sort of

then

you know people being really happy that
they've won something to them the heart

i mean it just i think what i like about

it is while it shows that touch yeah

it's a vehicle for for change but it

affects a lot of different things

and it's things that we might not yeah

when we're examining someone in clinic

we're not thinking like

is this you know rewarding or are we

making the bird song with the two birds

kind of

um doing it it's something that we

we innately do i don't think we

specifically always focus on

that all the different effects and i think it's nice to see an overview of touch is not so simple i think it shows very clearly that touch is not just touch um you might it's not just you touch someone you are affecting lots of different things or can affect lots of different things depending on how you use it um so it's just one vehicle for change but it can affect a lot of um different things um i liked doing it to show all just how varied um it can be and how it can be manipulated for lots of different things um the second one again like george said it's not so sort of scientific it does i think it it shows very well that touch is used through the sort of holding of hands in the middle but that's not the main thing um it shows i like how um we tried to show the different sort of aspects of each other so what the sort of practitioner might be thinking towards what the patient could interpret the same thing as very different so you see um with like b they're trying to help and console and c they're trying to move around but then on the other side you've got like g

where you know maybe they think that
the practitioner might be like yeah
thinking oh okay this injury you know in
a few weeks time they'll be getting back
on their feet they're getting
strengthening we can start doing rehab
but really actually the other one could
be thinking gee like oh no i'm gonna be
stuck in a wheelchair forever um
the same thing i like how it shows the
same thing can be interpreted in many
different ways depending on the prize
and everything that we've been talking
about it's not
the the communication the way things are
interpreted can be interpreted things
very differently with the same stimulus
um and i think that's what we try to
show
or hopefully we try to show that
active inference is a lot more
inclusive of lots of different things
that might be going on in treatment
um rather than just
one thing that you know just get them
better it's not
not not so simple there's a lot of
things going on that we've
tried to include
well on the theme of the same stimuli or
observation resulting in different
implications for different individuals
when i saw the handshake and the symbol
in the middle i thought of the dna
double helix and of like a sort of
feeling touched because it's more like a
brainwave synchrony
telepathy situation

otherwise their bodies would have been
touching so i think that's an awesome
conversation this was a great
um addition to the work itself
and we're really looking forward to
uh 44.2

next week at the same time so we'll look
forward to seeing everybody then again
thank you very much
see ya thank you
you